Profiling in the Age of AI – Part 1

By Yves Poullet

SUMMARY KEYWORDS
profiling, data, system, risk, individual, ai, people, algorithms, manipulation, predict, processing, services, problem, analyze, decision, groups, instance, functioning, important, artificial intelligence

00:14
First of all I would like to thank Professor Stefaan Verhulst for his invitation, I am very glad to share with you certain reflections I have about profiling activities. I think profiling development is quite crucial in our digital society, it puts at risk collective and individual liberties. So, thanks a lot for inviting me, Professor Verhulst. Just a few words about myself. My name is Yves Poullet and I am Emeritus Professor at the University of Namur, I have been the rector during eight years. I would like to add that I am now co-chairman of the NADI Institute. Now, the Institute is interdisciplinary research institute of the University of Namur, is composed by a certain number of researcher coming from different disciplines. It is very important for us to combine different disciplines to cross together different disciplines. I mean, lawyers, as I am, ethical people, definitely also come to scientists, managers, sociologists, and all people who are quite interested not only by the technical aspect, but also by the impact of IT on our society. I am also, as you see, Associate Professor at the Catholic University of Lille, and I am a member of the Royal Academy of Belgium. I'm also, and it is quite important in that context, I am also chairman of the working group Infoethics, the IFAP (Information for All Programme), as developed by UNESCO for analyzing the ethical problems caused by our digital society. So that's all. We start, if you want, and first the table of contents. Profiling activities are developing in all spheres of activity. It might be the administration wishing to better track down offenders, including potential offenders; or to offer more adequate services to citizens. It might be companies that want to know their current or future customer better, offer them new services or better select their employees. It might be hospital or medical scientific research, institutes fighting against diseases or using the technology for achieving better care.

03:25
More than 10 years ago, the Council of Europe enacted a recommendation taking into account the first profiling system. First profiling system based on traditional expert systems quite limited in their capacity and based on intrinsic logic, quite transparent to the different technologies - Internet of Things, big data and artificial mutation supplier together are modifying radically the methods and impact of profiling activities. AI systems often operate with potential bias and errors They operate in an opaque fashion combining the functioning of multiple neural networks. They rely on random statistical combination taking into account millions of data items and now make possible, unprecedentent, efficient and predictive profiling of individual behavior. That reality and the possibility and the possible lack of human mastership of the technology increase the risk incurred individually by the data subject, but also collectively by groups of individuals and even by society and our democracy. These regions militate in

Transcribed by https://otter.ai

favour of rethought and renewed legal framework for profiling. What I would like to do is to develop ideas for this new regulatory framework, taking surely into account the European data protection text, I mean, the GDPR and the Council of Europe Convention 108 plus. Firstly, I would like to sketch the context why AI profiling are so different from previous methods of profiling, notably the traditional expert systems. Which actors are involved within the conception, exploitation and development of the AI system used for profiling activities? Secondly, I will envisage and detail the main risks linked to that new technologies, as we regard this risk, we will particularly pinpoint new threats to our privacy. So, the risk of manipulation, of stigmatization, of discrimination, which are not only suffered by individuals, but also collectively by groups, and affects the development of our democracy. So, I will argue that a new regulation is to take into account not only the increasing risk incurred by individuals, but beyond doubt, those also the collective risk.

06:34
Recent EU text definitely envisioned these questions and I would like to quote this text. According presently to this recent EU text, we propose to distinguish among profiling activities, especially the higher risk profiling system, and that according to different criteria and to see how to face that. The third part of my presentation is the confrontation of the data protection area of the text, and the requirements of an effective data subject protection against the risk caused by profiling activities, underlining the problems, the application of this text meet due to the AI systems specificities, therefore, I will have suggest notably better attention to the role of certain actors, the enlargement of the duty of information, the right to have explanation in case of decision based on profiling, the right of recourse, new role for the supervisory Data Protection Authority and overall, the obligation now in case of higher risk profiling to proceed to a preventive assessment of the risk.

08:07
Furthermore, in order to cover the collective risk created by the system, I will refer to the solution proposed by a recent EU Parliament text about the ethical aspects dated from October 20 last year. First point now, sketch the context. Perhaps, it would be useful to start with definition of terms which definitively are important for the rest of my development. First, the term of profiling refers to any form of automated processing of personal data, including use of machine learning, consisting of the use of data to evaluate certain personal aspects relating to an individual, in particular, to analyze and/or predict aspects concerning that person's performance at work, economic situation, health, personal preference, interest, reliability, behavior, location, or movements. Traditionally, the profiling activities were based on profile, I mean, a set of characteristics attributed to an individual, characterizing a category of individuals, or intended to be applied to an individual. For example, the fraud suspect category, focus on the abstract category, directly presented by individuals who may have committed such a crime. The profiling aspects designate within an application pursuing a defined purpose, the application of the profile. The term profiling, in the first Council of Europe recommendation, dated from 2010 hanged on the sole term of profiling. Profiling means an automatic data processing technique that consists of applying a profile to an individual, particularly in order to take decision concerning her or him of one rising or predicting her or his personal preferences, behaviors and attitude. The term profile was and still is meaningful in systems which distinguish operation creating profiles from those that apply them, as would be the case in the definition of an individual employee profile or potential criminal. So,

Transcribed by https://otter.ai

the criteria used in profiling are transparent and might be discussed. Presently, with AI systems this distinction between the elaboration of profiles and the application is no more valid in the light of analysis of the functioning of artificial intelligence system, I prefer, as we have suggested for the new recommendation of profiling in preparation by the Council of Europe, when we are discussing such system to use the term model rather than profile. A model is, as you see, a mathematical abstraction, providing a simplified description of data to resolve the task at hand. It means the formula with the best parameters value for the solution. A model is not fixed, but evolves as it encounters more data, it reflects the fact that profiling we choose automatic learning matters does not function through causalism with the use of claims to the use rule formulated by expert a priori, but rather, on the basis of purely statistical and evaluative correlation between a large number of data. The pattern followed in the case of artificial intelligence is not causal explanation, but purely statistical thin findings. That distinction between profile and model is very important, since AI system are functioning on the pathway at least partially, I give an example, a very simple example. That's the problem of a nested agency, we want to predict the price of a property according to a certain number of criteria, but we don't know exactly the weight to afford to each of this criteria. You see, price is definitely calculated following parameters, surface area, the number of bedrooms, problem of shops, a proximity, problem of pollution and so and so, the age of the building. The value to be afforded to each of this parameters is not fixed at a priori, it will be fixed through statistical methods using definitively the sales already down in you know this rate or in a town in order to calculate seamlessly the wait to be afforded to all of this, to each of these criteria.

14:20
The methods of processing data in AI system. First, I would like to define what is AI. AI, and I take again, the definition given by the EU Parliament resolution dated from October 20 of 2020. So, it is very recent. AI means a system that is either software-based or embedded in hardware devices, you know the fabulous robot, I mean, you might think about definitely the automatic cars, drones, connected loudspeaker, car robots, and all things like that. So that software or the software is embedded in another device that displays intelligent behavior, that's quite important. The robot or the AI system will accomplish actions which normally are reserved to human beings. The third element is the fact that they are collecting, processing, analyzing and interpreting its environment. And apart from that, to take actions, that's very important, we will come back on that point, and they are collecting AI system are based on a collection of data very large collection of data, it means definitely big data, we will come back on that issue definitely rapidly, with some degree of autonomy and this degree might vary and we distinguish in the perspective different categories of AI system from supervised AI system, till deep learning system. I can make also that point. And finally, this AI system is taking action in order to achieve specific goals and I give a certain examples - credit rating, workers' evaluations, selection of course of customers. What is important in that definition is the fact that the EU Parliament insists about the importance of analyzing AI system in the larger, in the broader context, taking into account surely the connected technologies, it means that the technology of collecting data and notably the problem of Internet of Things, and definitely also the big data that means this vast reservoir of data by which the statistical calculation will be possible.

17:32

Transcribed by https://otter.ai

The methods of processing data in AI systems - there are a certain number of methods and I think it's quite important to see all these methods are used in profiling activities. Before doing that, I start with the definition of machine learning processing. Machine learning processing means processing using particular methods of artificial intelligence based on statistical approach to give computers the ability to learn from data, it means to improve their performance in solving tasks without being explicitly programmed for each of them. Let us come back precisely to the methods used by AI systems, I mean the different methods of machine learning systems. Profiling uses the data available for a person in order to better understand them or infer other information. But one, if we know first method, if we know in advance what purpose profiling is supposed to serve, and we have example of correct response for a sufficiently large sample of people, supervised machine learning will be used. A supervised algorithm is geared to learning the link between the data already available for a person and what the profiling entity wants to be able to say about the person. For instance, in a psychological study carried out on several volunteers, we might for example, look for the link between the data differed via questioner and their level of stress at work. For this kind of profiling the purpose is clearly established to supervised machine learning algorithm will be able to use all the questionnaires obtained and see how to best predict stress level at work based solely on the responses given by each individual. System geared to recommendation based on profiling may also use machine learning. This entails predicting individuals' preferences on the basis of other people's preferences, as commonly done on online purchasing platforms or in targeted advertising. Collaborative filtering is a recommendation technique, which compares seamlessly the consumer histories of the different users of a service on the assumption that people with similar habits will be receptive to similar products that they have not yet consumed. Numerous variants exist. In some profiling scenarios, it is sometimes difficult, if not impossible, to fully specify the purpose in advance as a response experiment expected by the profiling system is not known. This is typically the case when the aim is to segment the population of a country, passion community, of a hospital, or a company's customers. Groups of people will be formed automatically by algorithm, the point being precisely to learn new things about the citizens, patients, customers concerned in order to have a better understanding of them, to develop the service better tailored to them, to identify sub population at risk. The unsupervised profiling is very common and raises a question to what extent it is possible or desireable to precisely define the purpose of profiling. Profiling of this kind is generally intended to explore the data and isolate new knowledge that will then be exploited by humans, offten in a preliminary phase.

22:10

Another form of profiling where the expected behavior is difficult to predict is to have what we call anomaly detection, which involves detecting outliers within the population, exempli gratia individuals that are significantly different from the rest of the population, someone who will use a service in an abnormal way might be detected by instance in the case of fraud or quite simply, someone who should be ruled out of other analyses as they would skew or distort the results. Finally, there are also intermediary solution in profiling. For example, a semi-supervised algorithm can be used to carry out supervised profiling, even if the expected response is known only for a limited number of individuals. This means that a large population of individuals can be used to build a classification model, even through the correct categories now, only for low percentage of them. This is a common scenario when

Transcribed by https://otter.ai

the data are easy to acquire. But the response is costly and difficult to obtain, particularly if it requires the intervention of human specialists. This is the case for image processing, for instance.

23:55
In conclusion, there are many types of algorithms for profiling. Some of those algorithms will themselves help the designers to better understand the people they are studying and more closely define the profile they're aiming for. It appears important to take account of this variety in the fact that it is particularly difficult to predict what personal data will be useful in the case particularly of certain supervised algorithms. It must also be noted that the aforementioned algorithms are capable of processing not only digitized information, but also image, song, text, text sequences, and so on. Second point after the metals the condition. Anyway, AI system functioning principly use of big data, I mean a large collection of data, not necessarily store it in a unique place, but perhaps in a large number of database, but interconnected. Data which are not necessarily personal data, they can use anonymous data like for instance, a rich revenue of people living in a district, or the number of statistics about the number of terrorists belonging to a racial or ethnic population. The data are not necessarily sensitive data by nature. But at the contrary, they often reveal data, for instance, the geolocation data, the consumer habits. That need to make recourse to big data in order to have credible statistical inference explains the fact that certain players which have at their disposal a large number of data, have greater facilities for exploiting them for their own profit, and definitely for selling data, or their profiling service to companies through API. API Application Programming deficits among displayers definitely, we find the very large platform which collect a lot of data through the multiple servers determined by themselves or by their subsidiary or sister company for instance, Facebook and WhatsApp. The absence of EU large platforms creates a problem for EU market depending from the data collected mostly by non-EU companies. That explains the recent initiatives taken by EU authorities in favor of data sharing between company of the same sector for example, the sector of energy, the sector of the car industry, the health sector, and that in order to constitute typically you open big data.

27:23
Last point, the multiplicity of actors. The scenario, focus on the technical aspects of machine learning, but in addition to artificial intelligence specialists, many other people from different disciplines will also become involved. Profiling is by nature multidisciplinary as it requires input from specialists in database, software inventory, engineering, man-machine interface, lawyers, ethical people, etc. The infrastructure to store the data have to be designed, the software system for profiling have to be with great care constructed in a process, beginning with the analysis of the needs, and culminating in system release. The interface for exploiting profiling reasons must be designed, it must be ensured that the profiling system complies with legislation on personal data among others, checked that the data profiling system contains no bias and does not discriminate against any category of individuals. Furthermore, the setting up of an AI system involves multiple players having different roles. Indeed data can be derived from various sources, each with his own controllers, while the data sets on which that algorithm will already be running, may come from somewhere else. Some basic algorithms are available on the web in open source, while others purchased from their designers and the private licenses. The task of adapting these different components to the needs of a given sector or company is often performed by specialized players. Last but not least, the task of applying profiling to individuals or groups may be

Transcribed by https://otter.ai

trusted to a third party, usually, as a turnkey service provided generally by a large platform. All these players must be taken into account in our discussion about profiling regulation, as their contribution is vital. There are many possible configurations of players in profiling; within those configurations, each of these players bears a certain responsibility and has a different impact on the outcome. For instance, the producer of data might have not updated data or only have collected partial data, which led to bias during the application of the AI system or the algorithm purchased by quarterbacks.

30:47
I come now, to the second part, the second part of my speech, advantage and risk linked to profiling and typology of profiling applications. The term profiling brings together values operation that may pursue a number of purposes and present very different degrees of risk. Each and every one of us profiles other people at a certain point. We all try to categorize those around us on the basis of their personal attributes, whether relevant, objective, permanent or not. In short, we use the subjective/objective data in our possession to categorize others and infer without doubt with some margin of error in our assessment, other traits, our tendency of which we know nothing. Profiling is therefore in the nature of any human being, as we all try to put a label so that we can get a better handle of them and behave accordingly. However, using information to system changes the ways and scope of profiling and that for various reasons.

32:19
The first is that present-day information systems, through their activity and own presence, make it possible to increase exponentially, so the amount of data gathered. Data storage and communication capacity were previously limited today. Firstly, they've become pretty well in the finals as demonstrated by the phenomenon of big data. And secondly, the Internet of Things and the multiplication of services are so, that just a mouse click makes it possible to capture increasingly trivial aspects of our everyday life. Whoever is in possession of this data will be able to profile the individual concerned in ever closer detail. The second characteristic is, the use of the second reason, is the use of ever more powerful algorithms to analyze that quantity of data. 20 years ago profilers used the help of algorithms based on patterns replicating human rezoning, creating what were known as expert systems capable of standing in the data control. In that, they automatically transposed and applied the rules set up by human experts on the basis of their experience, the system guided reasoning and avoided the subjectivity and risk of discrimination that any human decision maker would have. These expert systems, which use totally transparent algorithms are now being superseded by what we call machine learning systema or deep learning systems capable of working on far more data than could be processed by experts. There is a system around correlation between expanding volumes of data using a large quantity of algorithms which fit on the data and come through it and gradually refine themselves accordingly. The variety and complexity of the model followed and developed by these algorithms interconnected are such that their function loses some of its transparency, even for the people who develop them or use them. Notwithstanding this previous consideration, it is quite clear that AI profiling systems are emitting a greater and greater success for obvious advantages. And I would like to describe quite rapidly these advantages.

35:46

Transcribed by https://otter.ai

The advantages of profiling are quite clear for corporations and administrations, which use such a system. For corporations, it is all about optimizing their action and investment, profiling will enable corporations to target their clientele, data mine their strategy using any number of parameters, better understand the reaction of a given community and so forth and so on. It is also an aid for making the right choice. For instance as we get the location of a subsidiary or for employees' recruitment or promotion. Furthermore, profiling is important for companies as regards security of their operation, it will help them to detect possible cyber attacks, fraud and so on. Last point, the use of AI system is also a good way to argue of the objectivity of the decision taken by the corporation. We all know the criticism of the profiling incentive carried out by a corporate manager was to recruit an employee or decide on a promotion, you mind subjectivity, bad mood, suspicion of the Stitcher app, and lack of quality and quantity of data serving as the basis for the decision will all be sources of doubt hanging over the decision in the eyes of some. Inversely, there is little to criticize in a decision proposed or provided by a machine having worked on plentiful data that appear to be objective and apply it without discrimination to all the candidates. If I take the example of recruitment, it is quite clear that we will analyze a certain number of objective data like for example, hand writing sample, statistic on a given category of candidates in relation to their studies and curriculum vitae, their behavior during the interview analyzed by facial recognition system, and affective computing technology. The neutrality of the working of the information system, the volume of data processed, and the apparent objectivity are obvious advantages in replacing human assessment with a digital one. For the public authorities, the same reason optimizations, security and object deviation are applicable. Furthermore, we're underlying that with AI profiling system public authorities enjoy a better grasp of the real situation and then might define better framing public authority action strategy in areas such as policy or employment, fighting crime, or definition of education program. Administrations also see profiling as a means of more effectively applying their regulation, notably by fighting against social insecurity or fiscal approach.

39:51

And finally, for an individual, we found the advantages for an individual with being profiled, the advantages are equally well-known. One example is the health sector analyzes by artificial intelligence of a patient, clinical antecedents and tumour tissues cross-referenced with those of 1000s of other patients will guide the doctor towards a given course of action in the space of a few seconds and predict an operation, sense of success. A second example relates to benefits for consumers. Someone wishing to buy a lawn mower best-suited to their needs and faced with a highly diverse market offers can be gradually guided by an attractive decisionating system towards a search engines that will provide also, this is about optimizing consumer choice. A third example is to be formed in the service provided by a music or film platform. Many of us are very grateful to platform operators for guiding us towards music matching our tastes that we did not know even existed.

41:22

But we also have to see the risk, the risk linked with the use of AI systems for profiling. The notion of risk is central, and we have to establish what risk we are talking about. First, we will analyze the internal risk, I mean the risk linked with the conception and the development, the technical development of the AI system. After that, I will analyze the individual's risk, I mean, the danger to our individual freedoms. And thirdly, I will analyze the collective risk faced by society and its democratic functioning risk, which recently have been emphasized by different texts enacted by the Council of Europe and the EU authorities.

42:29

Firstly, internal risk, I refer to the EU high-level group of experts report dated from April 2020 about what they call a trustworthy AI. The internal risks are very various. First, I think it is quite important to underline the risk of bias, bias means any prejudiced personal or social perception of a person or of group of persons on the basis of their personal traits. I take an example, it is the case of the testing of a system of facial recognition which has been made only on male Caucasian people. During the exploitation, we take conscious that this system which function ideally for white people was unable to detect different people who are black or who are coming from ASEAN countries - that's typically a bias. We must be sure that we are taking into account all the data which are necessary for dealing with the problem we want to solve with the AI system. The second problem is the errors in all IT activities - there are risks of error in programming, and as regard AI system it is also the case. The problem of opacity has already been developed. It is quite clear that when we have a large number of neural networks interconnected and working on thousands, on millions of data, we never know which kind of aggregation are made by the AI system, and thus it creates a problem of opacity of the functioning of the system and the difficulty to explain why the decision has been undertaken with regard to particular case. Fragility, fragility - that's very important, that is quite clear that the AI system might be fragile with regards to internal or external attacks, it has been demonstrated that the addition of a certain number of pixels with regard to facial recognition system will create a certain number of disturbations and a certain number of errors and that creates definitely the obligation to have a very secure system against internal and external attacks. And finally, evolutivity, when we have achieved the test of the first application, we must have a marvelous system working perfectly, but after, with new data, the system might evoluate because they are making new aggregations and they are making new links between new data and so, the system might evoluate in a non-wished way. The problem of evolutivity means that you must have a continuous attention to the performance of the system, not only at the beginning, but also in the course of its functioning. In this regard, the risk, the main risk is definitely the fact that these systems are not only detecting what a person is, but also what this person will do in the future, these systems are predicting systems and this possibility of prediction is definitely a very, very big risk. I have listed here a certain number of risks incurred by individuals and this list is quite impressive, I start with the beginning - the risk of rejection. It is proving increasingly easy to gather data as the source multiplied by omniplan technology. Storage costs have plunged and transmission capacity facilitates access to the data reservoir, the format or the concentration. All these explain the multiplication of big data and the possibility of their use by artificial intelligence. Within this data warehouse, individuals are focused on not this person, but from one part through the aggregation of a certain amount of data concerning them all this data regarded as little pieces of truths about each and every one of us, in that they represent a snapshot of our life, my presence in a given place, my Sir, my surfing what when my purchase of given products, my energy consumption, my dialogue recorded by connected speakers - all this data are revealing myself, but in the same time I am reduced to his data. From the other part we are focused not as person, but as data by the fact that the correlation of all the stored data with similar data collected from other individuals' anonymous data, specific to the entities or groups to which I belong, are more important than my specificity. So I am data because I am only analyzed through the data and that's the

Transcribed by https://otter.ai

data I am originating and secondly, I'm more and more not visit the such as person, but in comparison with other person. So the truth reflected by data grabbed from librarians are further component by statistic with data relative to other persons. It is difficult to deny that someone might be a con man, the perfect candidate or a person attracted by certain advertising or political party, if their profile or rather the model used shows that 95% of people with the same profile have the same potential or inclination. In short, in this alchemy of allegorithms the individual becomes an aggregate of data taken out of context and is ready to being nothing more.

50:53

Second risk - the risk of de-contextualization, the risk of taking data out of context, respecting context. The areas of trust in which a piece of data transmitted by the data subject is fundamental in our society, the use of the same platform, social network or searching drive in our various activities and presence of certain platform operators in various activities through their branches, enable this platform to cross-reference the data transmitted in different contexts, and so to break down the boundaries which individuals wish or may wish to lay down between the different areas of their life. For instance, I am acting as academic professor, but then I am also acting as a football player, I am also acting as a husband and a member of a fairy table association. All these sectors, I would like to maintain the separation between all these parst of my personality. The risk of stigmatization of computer memory is or at least seems unlimited, going well beyond the capability of your memory. It gives trace of our past, of certain events in our past which we would sometimes prefer to forget. This computer memory which is so useful for processing in the area of profiling raises the risk of people's stigmatization for their entire life. An insurer could keep the trace of a customer's illness indefinitely. A bank could keep a note of an unpaid credit. The risk of disintegration of the private sphere and the risk of unlimited surveillance. The fact that technology's area makes individual more and more transparent, as they are increasingly unable to leave unconnected to the tools and services of our digital society As we have seen technology records not only the trace, voluntary left by an individual on a social network of sites providing services and not only their movements in motion, as well as facial expression or location at any moment, through Alma via our other captors, we now ignore. Using the data captured through our eye system, and also by affective computing. This technology is designed to know or rather guess or predict our emotion sentiments. And I'm sorry, I come back to this phrase. So, the more and more data used by IT. For instance, they are using data through affective computing in order to guess or predict our emotional sentiments. We are using genetic data to better forecast or health future. So all this data are collected in our private and our public sphere and this distinction between public and private spheres is becoming more and more thorough. And it is quite clear that the protection of the private sphere, which was a mainstay of our rights, guaranteeing the inviability of our home as opposite to the public sphere, has now been done away. With the protection of when physical boots as inviable was traditionally viewed, and by the law too, as something that was fundamental for the construction of individual personality, the notion of home - a place where we could be free and out of other's site has also been turned upside down by technology developments that are driving the abolition of the distinction between public and private spheres.

56:06

Transcribed by https://otter.ai

The risk of manipulation, the opaque nature of the system functioning as another consequences the risk of manipulation and all the more so, as an artificial intelligence paves the way for what our colleague Antoinette Rouvroy calls algorithmic governmentality. What I want to say, as we have already emphasized, the profiles created constitute tools for analyzing not only the past, but also the future. The truth that these profiles claim to reflect is purely statistical and not exempt from bias. So, the profiles are valuable as a means of predicting future behavior, we can see obvious signs of this manipulation and what we are calling nudges. For example, the system, the AI system will suggest to a driver the best route to take; to a researcher how they can increase the hash index; to that of municipality the safe or no-go areas where policing is requested; to a minister of education or a teacher, the criteria whereby in theory, children could success at school; to a judge the risk of a perpetrator reoffending or the ruling, most closely keeping with the law, or rather, what has already been ruled as such; to read the books that they there are bound to enjoy, and so and so. Are all these manipulations reprehensible? Of course not. Sales men have always practiced bonus dolus without this being doubted as reprehensible. It might even be seen as a benefit for future customers by giving them more information, or even meeting their needs for guidance in a market that is ever more complex, that an ever wider range of product. Manipulation is reprehensible only if it represents an abuse of circumstances. I mean, that's a definition given by the Belgian legislation and manifests imbalance between the service provided as a result of one party abusing the circumstances linked to the position of weakness of the other party. Manipulation is punishable if it constitutes an abuse of the weakness of other people. Manipulation can have far greater ramifications when the profiling system is used for political ads in order to target voters with the right message, which may aim convincing them to vote for a given cause. The Cambridge Analytica scandal showed that this may be possible. Here the risk is not so much individual as rather collective in that it touched on our notion of democracy.

59:56

Risk of dehumanization, the ability to reason and take decision for oneself that that men could intrust to machines could result in replacing human reasoning forged by dialogue and consideration of other with an automated mechanism. This gave rise to major ethical concern. As can be seen from preparatory work, Europe's lawmakers have indeed become worried about this kind of automation the register to played by people by human people. And ultimately, by the human being, it has been a question of balancing from one part the risk of human decision making with everything that comes ith it, admittedly subjectivity, overemphasizing, poor vision in the decision process, and from the other part, the advantage of human decision making with its possibility of correction, dialogue and giving results. A further concern is the fact that the galloping automation of decision making process is engendering quasi-automatic acceptance of those decisions as valid and pertinent and in turn, divestment and abdication of responsibility by human decision makers.

1:01:45

Last but not least, the risk of discrimination. Finally, the possibility of automated reasoning being tempted by what is conventionally known as bias, has long been argued. This bias whether deliberate or not, may result in discrimination. The excessive weighting given to one criterion, the fact that such a criterion may conceal another criterion of discriminating against a category of the community - black people, women, for instance, disabled people, poor people, or, you know, in other cases, the refusal to

Transcribed by https://otter.ai

take into consideration a contextual element and specifically the data subject, making them a victim of the computer automaticity. The situation constitutes a final risk, both individual and where applicable to an entire groups a collective risk, and AI system analyzes of districts likely to harbour criminal casts version, not only in individual terms of the actual people living in that district, but also the district itself and its image, and will trigger social consequences. People find this quite clear that people deserting the district are refusing to go and live in there, or possibly increased police surveillance. Therefore, this is very much a risk - the risk of discrimination is exacerbated by the fact that the functioning of the decision making appear to be neutral and objective and the opaqueness of that function prevents the deciphering of the so-called logic followed.

1:03:54

After these individual risks, I would like to detail some collective and societal risks the AI system have raised more and more. I mean by societal risk, social justice groups' privacy, enbalement, cultural diversity, and democracy, effectively are integral risks the only ones evolved. Many of the risks mentioned above - discrimination, stigmatization undermine not only the freedoms of every individual as such, but also those of groups, be they ethnic, philosophical, low-income district residents and so forth and so on. Processing of this kind affects other values, particularly social justice or cultural diversity among individuals or among groups. And beyond that, and sometimes substantially the functioning of our society and our democracy, in particular, the abusive manipulation of individuals and the mind boosts freedom, but also human dignity in the conscience sense of the term. But why it is a danger to each and every one of us in this respect, it may also affect an entire population or have blanket effects by shaping the political views of citizens for example. Personalizing offers or services and excluding certain individuals from the potential benefits of the service have ramifications beyond individuals and impact groups of people, raising questions of social justice. Another thought in that context, by definition, big data drags us or big data drags us all, into a shared adventure. When my surfing data are gathered by my favorite search engine, and spread up by a vast database, to gather with millions of data items relating to the choice of other web users, the model that will at some point take a decision regarding me will clearly do so, in relation to all the data gathered, and by the differentiation induced by algorithm search between my data and those of others. Furthermore, there is also a domino effect whereby a person's choice to log on to a social network for example, will ultimately have a major influence on the choice of their friends and family.

1:07:14

With regard to the societal impact AI system might have on collective interest. Let us take the example of one-to-one insurance, civil insurance of civil liability for car drivers. Thanks to AI systems, coupled with the analysis of data received by sensors, placed within the car, car insurance are now in the position to propose premiums calculated as close as possible to the risk associated with the personality of the motorist and the characteristic of his car. I can imagine that I personally will have interest to be so profiled and will consent easily to place a sensor in my car, to take benefit of the system. But from a societal point of view, that one-to-one insurance puts the sacrosanct principle of risk fairing, a mainstay of our insurance system under huge threat. It is quite remarkable that traditional protection, that the traditional data protection of privacy issues, in the narrower sense of the term are transcendent in this way. We could say the same thing of an artificial intelligence system intended to predict the chances of

Transcribed by https://otter.ai

children doing well at school; or the risk of domestic violence against children who is in the population that would give weighing to certain data. The risk posed by this independent indentification are not just individual, but they are collective to us, there is a danger of stigmatizing certain types of communities. The necessity of broadening the concern beyond those which are targeted by the data protection legislation has to be envisaged. Different recent texts are enlarging to the ethical debates this collective risk and they have broadened the spectrum of risks to be taken into account. I quote the OECD recommendation adopted by the OECD Ministerial Committee in May 2019. AI actors should respect the rule of law, human rights and democratic values. This includes freedom, dignity and autonomy, privacy and data protection, non-discrimination, and equality, diversity, fairness, social justice, and internationally recognized labour rights. UNESCO is doing the same by mandating an expert group for drafting an international convention on ethical aspects of AI. The Council of Europe has nominated an expert group on AI and ethics, this is a group called CAHAI in order to do the same. Definitely through ethical aspects we are going broadly beyond the data protection and the individual liberties concerns. I quote here the EU Parliament's resolution dated from October 2020. This European Parliament resolution is proposing a regulation on ethical aspects, which developed the idea of AI actors as socially responsible, and they are socially responsible, not only with regard to individual liberties, but also as you see, in point A, they have to survey the development and deployment of the artificial intelligence system in order that this system contributes to improving individual development, collective well-being and the healthy functioning of our democracy. In point B, the problem of egality of people is underlined. And definitely point C, you'll see that they have to encourage the digital skills of the population and the promotion of multilingualisms as they have to also to promote a gender balanced manner that narrows the gender gap by providing equal opportunities for everybody. That's all this concern, definitely very important. And if you ever look at the recent draft of the Council of Europe recommendation on profiling, you will find exactly the same, the same concerns, you'll see that we have not only to fight against individual risks, and especially the risk of manipulation, and we have to promote the autonomy of human intervention. And we'll see that after in the last part of the speech, when we will have a look at the regulation of profiling, but also as you see the recommendations are emphasizing the collective risk. Now you see the principle of non-discrimination, the imperative for social justice, cultural diversity, and democracy. And you'll see it is a main point that profiling processing must contribute both to the well-being of individuals, but also to the development of an inclusive democratic system and sustainable society. We will come back on the other points of this very important text.

1:14:55

I come now to do a short classification of profiling activities. The first question is definitely the idea, the principle behind this classification. The principle behind this classification might be defined as a principle for a regulatory proportionality. All profiling activity do not present the same degree of risk and do not justify the same legal requirements according to the purposes pursued. Members that should ensure, I quote the text of the recommendation of the Council of Europe, the draft recommendation of the Council of Europe - member states should ensure that the regulation of profiling processing operation is kept proportionate to the purpose they pursue, to the nature and gravity of the risks incurred by the data subjects, the target groups and the general interest.

1:16:14

Transcribed by https://otter.ai

So, we might discuss, we might distinguish I'm sorry, we might distinguish different profiling activities. First, with regard to the purpose as we had the purpose they are pursuing. The first purpose is definitely the pre-contractual phase and the idea of having a targeted advertising. I would like to insist about the fact that more and more we are going from targeted advertising to exclusion and an adaptative pricing system. It is quite clear that targeted advertising was probably the first application that springs in mind when it comes to profiling. And one-to-one advertising seeks to match the personality and needs of the recipients as closely as possible in order to reach the consumer more efficiently. In many cases, the recipient of the message will find the chosen ads sent to him as more useful even when it is legitimate or wherever the manipulation that is interesting to all advertising is exponentially greater. As a message may be communicated in a covert, in the form of information more than advertisement, or in a misleading way. The recipient is unaware that the message comes from a third party and not from the website he is visiting. The manipulation will be all the more potent, furthermore, in that your profile will pinpoint your innermost self, playing on your emotion, inclination, including sexual winds, your race, even your disabilities, or opinions. The danger is particularly acute because in complex AI systems such as deep learning, it is difficult to make the models transparent and risk of manipulation is definitely hard to detect, prompting to talk of black box. Customer targeting can nevertheless have complimentary objectives. Adaptive pricing combines an estimate of the intensity of demand for a given product with the profile and offers different internet users a different price based on this variable. Another purpose is to select potential contracting parties by preventing some people from viewing certain messages or eliminating orders by making offers that are ridicilous. When I provide a case in the United States, Facebook was sued when lettings agensi used Facebook profiling services to screen out increase from prospective tenants or buyers belonging to certain groups - Africans, people with low incomes, LGBT and so on. In the field of employment too, it is uncommon for candidates to be selected on the basis of data, they have been analyzed by a machine including agency affective computing, and that according to criteria, which will have been articulated to a greater or lesser degree, when using the profiling system. Profiling, as we go in the contractual phase, I will be very rapid, the purpose of this profiling in this case will be to evaluate the partner, but this evaluation will itself have different objectives by being the evaluation of the contractual performance of an employee in connection with a promotion or dismissal. It might be the possibility to evaluate the credit value of a customer in the context of a bank trying to decide how much to lend or to charge in premiums, or to data mine customer profitability. Alternatively, there may be financial or other ways to be assessed when evaluating a partner, whether business or individuals in order to decide whether it is worth continuing the relationship.

1:21:48

Third types of profiling activities. There's the the marketing of data of profilings. Orders of big data may be tempted to offer profiling services to third party, either a simple provider of data or in response to request from the customer. In that case, they will develop the necessary algorithms and apply them to the data given by this customer. The big data holders will then make the person profiled available or offer to perform themselves whatever operation the third party requires. The existence of data sets whether open source or proprietary, or the provision of algorithms free of charge by research laboratories on a commercial basis by company offering profiling services were discussed already. The Facebook fan page after the Cambridge Analytical scandal are worth citing here. In the former Facebook provided administrators of NPR with information and profiles of people visiting the site, so that they could learn more about the visitors who like their fan page and design a better strategy to keep them. In the Cambridge Analytica case it concerned the use of data generated by the social network and supplemented by Cambridge Analytica co call it researchers. In this Facebook Cambridge Analytica scandal 87 million Facebook users were affected when Cambridge Analytica began arresting their personal data in 2014. The data were used to influence voting intention in favor of a politician who paid for Cambridge Analytica service. The aim was to ascertain people's political leanings and even predict how they will vote and to market the findings to political operators so that they could tailor their political marketing strategy. Just recently, the Financial Times revealed how Google had been sharing with commercial companies precisely for the purpose of facilitating their profiling activities.

1:24:52

Profiling may add to research, especially as we gather medical research. I think that everybody knows the interest for medical research to use AI systems for profiling patients and this to predict their future health and also to define the best, the best way to help people to maintain their health status. Profiling by public authorities, profiling for governments and administration, profiling alows public authorities administrations to pursue a variety of objectives. First and foremost, it is useful in devising strategies, whether economic expansion policy, or policy and I'm securitizing, housing mobility, education, employment assistance. To this end, authorities consider a myriad of factors and combine reams of data from public or private sources to produce predictive models for data mining or the impact of particle policy is likely to have. It is too easy to see how, if not programmed correctly, be as poor data quality error to the algorithm, activities of this kind could affect certain groups or at any rate how a decision based on such forecasts could affect the member of the groups. One area where AI and profiling systems can be a really sort of efficiency gains, is when it comes to implementing rules and regulations. As part of a philosophy of benevolent governance, where the state plays a proactive role with respect to its citizens. Such a system can be used not only to support, or even select people who could benefit from special assistance, or to ensure to the students to receive the best possible advice about education pathways and programs. They might also be used to detect problems within families. For example child abuse or even to find social security fraudsters or tax evaders. Predictive justice, which is intended to replace judges is also worth mentioning in this context, insofar as any dispute can be provided, according to the many and various characteristics of the case, and that is analyzed in the light of previous decisions. The second classification is more recent. That's the classification by the importance of the risk incurred by individuals or society. It is our belief that not all profiling should be regulated in the same way. Profiling needs to be regulated according to the gravity of the risk engendered and the application of the processing for the data subjects of society. That notion of higher-risk processing has been introduced for the first time by the general data protection regulation, to justify the obligation for data controllers using the system to conduct and attain privacy risk assessment. The EU Parliament's proposed regulation of ethical aspects of AI is also using the notion of higher-risk processing and that in a broader significance. First, they are using that not only as we got individual but also sociatal risk. Secondly, the annex of the proposed regulation defines the concept by defining an exhaustive list of the sector purposes which leads to the qualification of processing using AI systems as high-risk processing. This processing must be subject to a risk assessment conducted by external auditing company or directly by national ethics, supervisory committee.

1:30:20
As regards profiling activities, recently, the notion has been taken again, by the Council of Europe recommendation on profiling, you know, that this recommendations are still in discussion, but the draft of the recommendation defines the notion as follows: the expression 'high-risk profiling' refers to profiling processing operaions which operations entail legal effects or have a significant impact on the data subject or on the group of persons indentified by the processing. For instance, it might be the problem of terrorism risk, or definitely the problem of credit value, or the problem of education. First point, the second point is profiling, which because of the target public or the context of the purpose of profiling, particularly the possibility of abuse, of misuse of the possessing a room of information power, notably when it concerns minors and vulnerable people involve a risk of affecting or influencing the data subject we have here the fight against manipulation of data subject by profiling system. The third point, the problem of special categories of data, or having, for purpose to detect or predict then this, for instance, the possibility this the Cambridge Analytica example, the possibility to predict the opinion, political opinion of people. There's a possibility to deduce the health future of a certain number of patients through analysis of genetic data, for instance. And finally, the profiling affecting very large number of individuals including such processing carried out by online intermediary services for their own usage of third party usage. It is there the problem of the very large platforms, which have the possibility to collect a lot of data and offer to third party a certain number of profiling services. According to this draft recommendation, well, the envisioned profiling activities of high-risk nature, controller should notify its existence to the supervisory authority, and if requested by the latter, informed about the corrective measures taken on a visit and make the controller by the supervisory authority of a label. I will have to conclude this first chapter. It is quite obvious that artificial intelligence has substantially modified the functioning, the actors, and the risk of profiling. Modern profiling is no more necessarily linked with profiles in machine learning, many models do not explicitly manipulate profiles. They're working with model. Furthermore, moderm profiling are using big data complex algorithms, including deep learning systems, which multiplies risk of bias and errors. Modern profiling is based on the statistical aggregation of vast amount of data and no more on a transparent logic causation system.

1:34:40
Modern profiling is functioning apart from complex and unpredictable interaction of a large number of neural networks. The problem is opacity of their functioning. Modern profiling implies multiple actors whose roles and liability must be precised. Modern profiling amplifies the risks faced by individuals due in particular to its predictive capacities. But beyond this individual risk modern profiling creates collective risk and in particular risk of discrimination (group's privacy), according to unforeseeable criteria.

Transcribed by https://otter.ai

Profiling in the Age of AI – Part 2

By Yves Poullet

SUMMARY KEYWORDS
profiling, important, data, data protection legislation, processing, definitively, problem, system, regard, profiled, ai, subject, decision, purpose, order, people, case, person, risk, clear

00:01
Thanks a lot for joining me again, we have developed a certain number of definitions explaining our profiling system, our running, our development, the actors, the risks linked to this profiling system and definitively we have enunciated the fact that there are not only individual risks, but also societal risks. And definitely, I will come again to this question in the third part. And the third part will be dedicated to EU data protection text facing profiling systems using AI techniques. And immediately, I would like to pinpoint the two points I will analyze. The first one is definitely the fact that, as we have said data protection legislation are considering only individual risk. So, what about collective risk, perhaps we have to find a solution to address this collective risk beyond data protection legislation. That's what we will have to look in the first part of this presentation. Then the second point is perhaps more difficult. We are convinced that data protection legislation do present a certain number of shortcomings in order to address the individual risks each data subject is reporting. And what I would like to do is definitely to see these shortcomings and to see how it would be better to modify data protection legislation, EU data protection legislation in order to face this new risk. For addressing these two points, as I mentioned, I will refer to the report and the recommendations that Benoît Frenay and myself have addressed to the Council of Europe in the context of the preparation of a recommendation on profiling at an AI age.

02:49
Let's come now to the first problem - the need to go beyond data protection legislation. We are convinced that it is very important to enlarge the objective of the regulatory text in order to protect people in society against profiling risk. And I mentioned here the fact that thde EU resolution on ethical EU resolution of the Parliament, the EU parliament on ethical aspects are considering this enlargement of the objective as you see the respect for fundamental rights and freedoms, notably the right to privacy and the principle of non-discrimination, but also the imperatives of social justice, cultural diversity and of democracy shall be granted during the processing of personal data subject to the recommendation. That's very important. That's very important, then it implies as it is asserted in the third paragraph, it implies and the fact that the supervisory authorities competent with data protection legislation have to work more intensively with all the bodies, like definitely the commission for equal opportunities, like the commission created for ensuring consumer protection or definitely the authorities as regard competition and the promotion of media diversity. That's the first thing as we got no... I'm sorry, let's look in the other sense.

Transcribed by https://otter.ai

05:03
Forget the consequence of this enlargement, it is important for both EU Parliament resolution and for the recent recommendation on profiling drafted by Council of Europe to develop what we call a priori evaluation of the risk. And the idea, the main idea developed by the EU Parliament's version is to say that we need a continuous assessment of both individual and collective risk, and this continuous assessment must be done by qualified and adequately knowledgeable professionals and must take into account various impacts, including their legal, social, ethical and technical dimensions. It is the idea to have multidisciplinary evaluation of risk. As regard the problem of certain AI systems, notably, the problem of deep learning systems, it is quite clear that we have to ensure a certain number of quality in especially the problem of transparency, and the obligation to have, what we call, a human interface between the person who is subject to the decision and the decision maker. And finally, the idea developed in the third paragraph, the idea to develop an independent and qualified certification mechanism would be interesting, it is very, very interesting to have lable certification, which might be opposed to the different profiling systems in order to discriminate between certain software which are certified and which a priori creates a presumption of good development and the other one, which must be evaluated quite seriously by this organization, independent and multidisciplinary.

07:59
We come back now to the main point, I mean, the question of the EU text on data protection facing profiling system using AI techniques. I see a lot of shortcomings. And I will start with the problem of definition going after to the main principle of legitimacy as regard to processing and after that, we will see the rise of data subjects as regard this processing, definitively the obligation of security related to this system. The first question is a problem of definition, problem of definition. We have seen that AI profiling systems are using not only personal data, but they are using also other data than personal, I mean, and it's very important, anonymous data. For instance it is quite clear that if you have to calculate the credit value of a person, the bank will use not only personal data up to seven about all the data subject, but they will also use a certain number of anonymous statistics, for instance, the average revenue of a person living in a certain district, or the fact that this person living in this district has this kind of education competences. So I think that's very important to take into account anonymous data. Another point is a problem of legal entities, more and more large companies are using profiling and profiling systems for determining the credit value of a small and medium company, they are predicting the future of this company, seeing bankruptcy for example, in 3 years or at midterm. So, I think that if we want to consecrate the freedom of undertaking, it is very important that the profiling of small and medium enterprises would be regulated. It does not mean that you have to extend data protection legislation to legal entities, but it means that certain rights afforded to data subject, physical data subject, natural person must be or also given, as regard profiling systems to legal entities, I mean, definitely the right to inform it and the right of access, I think that's really important that the quality of the data will be assessed and so and so.

11:38
A third point, third point is the problem of the definition of sensitive data, traditionally sensitive data are defined by the nature as such of the data. It is quite clear that you have data if the data is related, if you have sensitive data, if the data is related to your health status, it is quite clear that data is sensitive, if it is about your racial origin. The problem with profiling is you might deduce from trivial data from a non-

Transcribed by https://otter.ai

sensitive data definitely, you might deduce sensitive data. I take an example it is quite clear that Cambridge Analytica was considering the data exchange within the Facebook network and they were not sensitive data, by nature, but apart from this analysis, the Cambridge Analytica was able to deduce the political opinion of this person. So, that's quite important to have a definition of sensitive data not by nature but by purpose of the processing. The second problem of definition is related to the actors. We have seen that in order to build up a profiling system you need a lot of people, of actors playing different roles. As we have seen, you have a certain number of actors who will furnish the data or they will furnish algorithms, or they will furnish, what we called turnkey application in order to determine the person to be profiled or the right profiles. So, you have a lot of people and it is not clear that all these actors will be qualified, if I have a look at the data protection legislation, as a data controller or data processor, a furnisher of data is not a data controller, is not a data processor, but it is obvious that his role might be decisive as regards certain errors, as regards certain bias. It might send data, which has not been updated, it might send only partial data. So, it is very important that we take these actors also into consideration, and that these persons have also a certain number of duties, for instance, or to give the documentation in order to give to the data controller, the documentation in order to control the quality of the data, in order to control the appropriateness of the data. As regards the purposes of the processing, that's really important. Another point is more specific to the platforms As we have seen, very large platforms, are very often offering a certain number of services, to give the data according to the requirements of their customer in order for their customers to profile people, or they're offering their profiling services to certain companies. In the decision against Facebook, the European Court of Justice, has decided that Facebook who has delivered a certain number of data about the profiles of the fan club, but the requirement of this fan club as a marvelous, marvelous block. And he would have liked it to have a better cognisance of their customer in order of having this better cognisance. He have asked to Facebook to give a profile of these customers and Facebook has done that, and Facebook has been considered as a joint controller because he was offering the means of the data processing and this is to be qualified as join controllers.

17:13

So, second category of problems, the problems as we get really messy and the proportionality of the processing. We have seen, and it is often the case with deep learning and unsupervised AI systems that very often the person who's using an AI system has no a priori knowledge of the purposes. He might attempt with the profiling, he has no knowledge about of the categories of people, he will discover with the profiling, he's thinking about for example, only targeting advertising for people and after that, he discovered that it would be quite interesting also to select people according to the possibility of adaptive pricing as we have seen. So, as we are profiling t is very difficult, when you are using an AI system to define a priori the purpose you will attend or you will achieve. And that's problematic. Why? Because the GDPR and the convention when under it as a fundamental principle, you need to fix a priori, the determinant and specific purposes and not to process that in a way incompatible with those purposes. So, according to this data protection text, you must fix since the departure, the purpose. In the case of use of AI systems, it is really difficult. So what we propose is to have a sort of permanent evaluation of the purpose and the obligation definitively to adapt automatically the information you give and definitely the rules you have to follow to this evolutive definition of your purposes. The second point, there is the question of consent. The consent is, according to the data position text, the first ground for legitimizing processing, but, when you are entering in the reality of profiling systems, it is

Transcribed by https://otter.ai

quite clear that due to the opaqueness and the impossibility to have a real choice, consent becomes a privacy bug. Everybody of us know that the simple click on 'I do accept' is the only way to manifest your concent in most of the websites you're visiting. And it is quite clear that this consenting is not informative, is not free, because you urge to have the service because you have no idea about all the purposes followed by the person and if you want to analyze all this, all this information, in certain case, it is possible to do that, you will lose, you will lose rapidly your passions and you will finally the 'I do accept' this. So, I think consent is no more the right way to legitimate most of the profiling processing, I have a certain number of solutions acting that in certain case regulation must intervene. You must have legislation, for instance with regard profiling processing used in the context of employment, employees' recruitment, you must have I think definitely regulation as regards certain insurance for bidding for, for instance, a certain number of techniques like one-to-one insurance or the use of a certain number of data as regarded as insurance.

22:40

Another solution, I think very important is definitively the fact that it will be very interesting to have collective negotiation like in consumer protection, in order to ensure the legitimacy of the processing I think will be quite interesting to have a discussion between the provider of the service and definitively the representative of the consumer, notably through consumer associations or civil society associations. The other thing is the fact that as regard large platforms, intermediary services like Google, like a certain Google search engine, like Facebook, social networks, it would be interesting to oblige this large platform to have by default, non-profiling service and definitely to be important for reinforcing the autonomy of people to have restriction of the right of data subject to oppose to the selling of your data to third party. I think that's very important. Very important. I agree that Spotify has my data but at the same time, I disagree with the fact that Spotify sells my profile to third companies.

24:37

Always as regard the problem of the legitimacy and proportianality of the processing. The last point and that's a very important point. You know, the data minimization principle, you may only process data only if they are pertinent for achieving the determined purpose pursued by the processing. That's where you have a problem when you are speaking about profiling systems especially as regards deep learning systems because the systems are using a lot of data without knowing, as we have said, without knowing exactly the relevance and keeping, and they are keeping them for potential future use. I think a stupid example perhaps in order to detect fiscal fraudsters I will discover with my AI system that dentists which have a red car are definitively suspicious people, they are at least suspected people. So, it is quite clear that it is very difficult a priori to determine which data is pertinent as regard the outcomes expected by the AI profiling processing. That's really a problem and I have no definitive solution, but what we have proposed is the following: the minimization principle must be nuanced in machine learning system, it is difficult to know a priori which data will allow significant correlation and moreover, it is important to limit the profiling processing to categories of data that the data subject can reasonably expect to be taken into account in view of the legitimate purpose of profiling. The reference to this principle of reasonable expectation of the data subject seems to me a more acceptable, a more acceptable criterion than nothing. That's what I have defended before the Consultative Committee of Convention 108. And as regard the duration of the storage, you see that the personal data used in the

Transcribed by https://otter.ai

context, should be stored in a form that allows the identification of the data subject for a period no longer than it is necessary for the purpose for which they are collected and processed.

28:08

I go now to the question raised by the issue of quality of data and algorithms and I go directly to the solution we have proposed, I think it's very important that there are appropriate measures in order to correct data inaccuracy factors and limit the risk of error and bias inherent in profiling and definitely it's very important that these measures are regularly started again. It is very important that we need to have an evolution, we need to take into account the evolutivity of the system, and thus that we have to assess regularly the quality of data and the risk of errors and bias. Finally, that's very important, we have already spoken about, it is important that the person who are furnishing data and algorithms are furnishing at the same time the documentation in order to avoid any bias or errors in the processing. Problems as regards the data subjects' rights. As we have data subjects rights, I think what we need is definitely an enlargement of a certain number of this data subjects rights, the enlargement of the information to be given. I think we have to give the results of the profiling processing, it is very important, and the purpose for doing so. I think it is very important that we create a data subject not only for individuals, but for categories of persons or bodies to whom or to which the personal data or the result of the profiling processing may be communicated. It is very important that when I am profiled by Google, I have to know exactly to whom Google will give these profiles or the data needed for building up new profiles by other companies, it is very, very important. The second point is definitely that it would be interesting to have when companies using profiling processing to use an icon fixed to the website, to the product is sent in order to inform the people that it is using a profiling processing system. And this icon may refer to definitely to web pages where a certain number of information will be given to the data subject in order to know better how the model functions or which possible impacts will be addressed by using this profiling system; the beneficiaries of the data, the categories of data used including anonymous data, that's very important. Enlargement of the right of access, we have just mentioned the fact that it would be definitely useful to give not only access to personal data collected about people, but also about anonymous data, which are used for the processing, and definitively that's also very important, the need to have explanation on the functioning of the AI system, the reasoning underlying the processing of his personal data and that was used to attribute the profiling to him. Other rights the data subject has at his disposal, his or her disposal, the right not to be idefined and profiled. We have already said and I think this is important, we have already said that certain intermediary services, in particular large platforms, would have to offer a service of non-profiled access to information about their services. When I want to get a better information of what is offered by Spotify or by a Google search giant or by other platforms, I think it will be absolutely needed that this access to the information about services would be anonymous, there is no reason for the platform to get information about the potential customers. So that's very, very important to have this non-identifying possibility of access.

34:38

The right to 'faithful' devices, we have said that a lot of data are collected through devices, Internet of Things devices, have already placed everywhere. It is very important that the person who is encountering this kind of device will be informed about the existence of this device, the fact that there is a device in the wall, the fact that there are devices on the product boxes, and everything like that, it's very important that there are devices in the clock or in their GSM, that's obvious. But more than that, I think that's very important that the devices are operating faithfully, it means that, if they are collecting data, it will be absolutely needed to know exactly which kind of data are collected, to whom this data are sent, or for whom they are collected, that's very important.

35:56

And finally, I think the right to free consent and that's, that's a very, very big concern. I take an example, you might imagine that Spotify is developing different services as Spotify did already, Spotify offers a service for people who want only to have a good library of music and to have the possibility to search their preferred music or the music they want to listen. But you might imagine that Spotify will have a supplementary service saying to their customer, if you want to receive the music according to your profile, that's not a problem, we can do that through definitely profiling processing. I think it's really important for the customer, to have the possibility to say I want to have only access to the library or I want to have access to library plus this profiled service. In the second case, it is quite clear that Spotify has a justification to use more data than in the first case

37:38

The right to oppose, we have already spoken about the possibility for people to oppose to the selling of their profiles, of their data and definitely to be quite interesting to have the possibility to oppose when the processing concerns the sensitive data - political, religious, philosophical or related to trade unions, prospecting, relating to trade unions, it is quite clear that in that case, the data subjects must have the right to oppose to this kind of processing, right, not to be subject to an automated decision. I think that's a very, very important topic. Why? Because profiling is often the condition for a company to take a decision, and the decision is definitively often dictated by the profiling. There are different things I would like to say there about it. The first, I think that the data protection legislation limits the principle to the possibility for data subject to not to be subject to a solely automated decision. It means that if a bank say okay, as regard the credit value I am using definitively an AI assistant, but be sure, that's only for helping my employees to decide. So, they say automated decision is only used for preparing the decision but the decision is taken by a human. I am not convinced by this distinction between the decision and draft decision. Why? It is psychologically, I think it's very difficult for a bank employee to say the automatic decision or the profiling system indicates me that this person has no credit value, but nevertheless I will afford, I will grant the credit requested. Why? Because it is quite clear that the bank employee will be afraid to be considered as liable for not having followed the advice given by the AI system. So, this distinction between decision or draft decision or proposal of decision seems to me not relevant. I think what is very important, I think what is very important, is when there is a decision or proposal of decision given by a profiling system definitely it would be needed that a human person will explain, not completely we know that as we are deep learning that is impossible, but as far as it is possible to explain in a clear and understandable language how the model is functioning and why the decision has been taken, and the same person has to listen to the objection proposed by the data subject to whom the decision is supposed, he has to listen to him and he must have the competence to say okay, I have listened to you and definitely I modify the proposal of decision I am for example, for instance, I am giving you the credit you have asked. And if it is not the case, the right to have recourse, the right to challenge the decision must be given before an internal body of the data controller, but

Transcribed by https://otter.ai

definitely also before the judges. I think that's very, very important to have this procedure put into place in order to avoid what we might call the technological imperative.

43:24
I come to another point - the question of data security. The first problem is definitively that there is with AI systems working on very large big data there is a possibility for disanonymization of the data included in the big data, it is possible. So, it is very, very important that the risk of re-identification of anonymous data would be taken into consideration by the person who is using an AI profiling system. As regard AI profiling systems, there are other precautions to be taken. First I think it's very important to have a robust and safe system meaning definitively the possibility to resist to external or internal attacks, as we have said, but also to detect errors and bias, and definitely have to ensure that the results of the system are reliable, and they are in conformity with the model and reproducible. We have to address the problem of transparency as we know, transparency as regards certain deep learning profiling systems would be impossible. But we have to ensure, that must be very clear, we must ensure the explainability and the traceability of the processing results. And that's very important, if we consider the fact that often the data controller will say, I don't want to give you any explanation about how the processing is functioning. Why? Because I have intellectual property rights, I have the problem of secrecy, and I will refuse. So I will refuse on this basis. As you see, we have proposed to the Council of Europe in the context of the recommendation, we have recommended to add a provision saying that the data controller will ensure that their intellectual property rights and trade secrets, are only minimally opposed and, in no way will they be able to oppose the request of a data subject or of a group to be able to understand the decisions or proposed decisions taken from the profiling processing operation.

47:04
Last point that's obvious - the need to have an interdisciplinary evaluation, and regularly.

47:15
I go directly to the conclusion, I'm afraid, I've spoken too long. But I would like to address to you certain conclusions of being that they might be of interest for you. First, that's the first point, profiling using AI technology definitely is one of the major issues for liberties, but also for the development of our democratic society. That's very important, not only individual risks, but also collective risks. The second point - data protection texts have shortcomings because they do not take collective and societal risk, but also because their concepts are not always adequate to cover the new realities. We have proposed as regard the collective risk to develop a new regulation, and I think that the EU Parliament resolution on ethical aspects might be interesting. As regards the fact that data protection texts are not always adequate to cover the new realities, we have made a certain number of proposal items, the need to have a preventive approach by summarizing all this proposal. First, I think that concent might not be a panacea. I think that it will be needed, in certain cases, to regulate by legislative intervention certain profiling processing, and definitively that in all the case, when a profiling is a high-risk system, we have to go through a risk assessment, multidisciplinary, multi-stakeholder procedure, led by external auditors, external and independent auditors under the control of public authorities. That's what I to say about this legal perspective on profiling and profiling techniques. I hope what I said might be interesting for you, and I hope that we might stay in discussion. You have my email, please go forward with this discussion through this way. Thanks a lot.

Transcribed by https://otter.ai